

Trace metal concentrations along tributary streams of historically mined areas, Lower Lahn and Dill River basins, central Germany

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ABSTRACT

Watersheds with a history of mining frequently have high concentrations of trace metals stored in downstream floodplain sediments. To assess the contribution of tributary streams to the elevated trace metal concentrations previously documented along the lower Dill and lower Lahn Rivers of central Germany, I examined the concentrations of Cu, Pb, and Zn in fourteen cores collected from near-channel sediments along 9 tributary streams. Five tributaries empty into the lower Dill River and 4 drain into the lower Lahn River. The tributary streams drain areas that were mined for Fe, Cu, and Pb for more than a century, with the last mine closing in 1983. Cores were also collected from near-channel sediments at 7 sites along the lower Lahn River to more broadly assess the storage of trace metals along the river. Large quantities of Cu, Pb, and Zn from mining activity are not stored in near-channel positions along the length of the sampled tributaries. Mean and median concentrations of Cu, Pb, and Zn along the tributaries are less than the mean and median concentrations along the lower Dill and Lahn Rivers; concentrations along the tributaries differ little from background trace metal levels and are less than contamination standards contained in the German Federal Soil Protection Act. The mean concentrations of Cu, Pb, and Zn along the lower Lahn River are greater than background concentrations and mean concentrations exceed contamination standards from the German Federal Soil Protection Act. Tributary streams are not a source for the elevated trace metal concentrations found along the lower Dill and Lahn Rivers and mined areas will likely not serve as a source of future heavy metal pollution. The results of this study indicate that the mining process did not release large quantities of metals to the fluvial system.

1. Introduction

Contamination of floodplain soils and alluvium with trace metals poses a long-term threat to the environment, especially when metals are stored in geomorphically sensitive positions such as channel banks and near-channel floodplains. Should the metals be remobilized during floods or channel shifting, metal-contaminated sediments can be conveyed downstream. Trace metals typically originate from anthropogenic activities such as industry, agriculture, and mining. In parts of western Europe, trace metals have been discharged to the natural environment for many centuries (Matschullat et al., 1997; Brännvall et al., 1999; Renberg et al., 2001). The peak discharge of metals to the environment is usually attributed to the period from the early- to mid-nineteenth century until the early 1970s (Renberg et al., 2001; Middelkoop, 2002; De Vleeschouwer et al., 2007; Ciszewski and Grygar, 2016; Dhivert et al., 2016). Mining activities have long been a potential source of trace metals to the fluvial system. In Europe, trace metal mining began as early as 3000 years ago (Monna et al., 2000), and numerous studies have chronicled mining dating back centuries

throughout western Europe (e.g., Hudson-Edwards et al., 1999; Mascaro et al., 2001; Coulthard and Macklin, 2003; Dennis et al., 2009; Hürkamp et al., 2009; Foulds et al., 2014; Resongles et al., 2014; Beane et al., 2016; Bird, 2016; Schillereff et al., 2016; Horák and Hejeman, 2016). The vast majority of the mines have been closed for many years, but contaminated sediments remain stored in channel banks and near-channel sediments. In these geomorphically active zones, contaminated sediments may be remobilized and carried downstream (James, 1989; Marcus, 1989; Leigh, 1997; Förstner, 2003; Foulds et al., 2014; Lecce and Pavlowsky, 2014; Clement et al., 2017).

In Europe, the impact of historical mining activity on rivers and streams has been extensively documented. Studies from the UK have shown that historical mining represents a significant source of contamination to the fluvial system (e.g., Hudson-Edwards et al., 1999; Coulthard and Macklin, 2003; Dennis et al., 2009; Beane et al., 2016; Bird, 2016). Individual flood events can transport substantial quantities of contaminated sediment downstream, impacting important agricultural land and potentially contributing to livestock poisoning and increased mortality (Foulds et al., 2014). Germany has a history of

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mining that extends back centuries. Some of the earliest activity has been traced to around 3500 yr BP in the Harz Mountains of central Germany (Matschullat et al., 1997; Monna et al., 2000). Other regions with a long history of mining include Rammelsberg (Matschullat et al., 1997; Brännvall et al., 1999; Bindler et al., 2011) and eastern Bavaria (Hürkamp et al., 2009; Raab et al., 2011). Early mining activity was relatively inefficient in extracting and processing materials, typically leaving behind large quantities of trace metals that could be conveyed to downstream floodplains. Today contaminants stored in floodplain sediments provide an “archive” of past mining activity and the resultant metal contamination (Hürkamp et al., 2009). Vertical trends in trace metals can be useful in documenting the movement of metal-contaminated sediments through drainage basins and assessing the potential for near-channel floodplain sediments to serve as future sources of downstream contamination (Gocht et al., 2001; Middelkoop, 2002; Vallius, 2014; Lecce and Pavlowsky, 2014).

Studies of the distribution of trace metals around and downstream from mines have established typical spatial patterns for metal contamination. Floodplain sediments and channel banks have been shown to contain elevated metal concentrations tens of kilometers downstream from former mines (Coulthard and Macklin, 2003; Dennis et al., 2009; Foulds et al., 2014; Lecce and Pavlowsky, 2014; Beane et al., 2016; Bird, 2016; Wadige et al., 2016; Clement et al., 2017). Metal concentrations typically decrease with distance from the former mines, reflecting the additions of cleaner sediments from downstream tributaries and the tendency for metal-contaminated sediments to be deposited in close proximity to the mine. Trace metals stored in floodplain sediments from prior mining activity are regarded as *legacy sediments* (James, 2013) that have the potential of serving as sources for future metal contamination (Leigh, 1994; Schwartz et al., 2006; Ciszewski and Turner, 2009; Dennis et al., 2009; Wyżga and Ciszewski, 2010; Foulds et al., 2014; Bird, 2016; Wadige et al., 2016).

I previously documented storage of trace metals in floodplain sediments and changes in metal concentrations over time along the Lahn River, a predominantly agricultural watershed of ~5000 km² in west-central Germany (Martin, 2009, 2012). The 25-km reach of the lower Lahn River between Wetzlar and Weilburg was found to have the highest metal concentrations (Martin, 2012). This was attributed to the input of contaminated sediments by the Dill River, a tributary that empties into the Lahn River at Wetzlar. To assess the potential contribution of trace metals from the Dill River to the lower Lahn River, I examined anthropogenic enrichment of trace metals in floodplain sediments along a 25-km reach of the lower Dill River, documenting that near-channel sediments are enriched with trace metals (Martin, 2015). The potential source of trace metals in the lower Dill River is either heavy industry in the upper Dill River basin (Fröhlich et al., 2008) or past mining activity (Meyer, 2007).

To assess the contribution of formerly mined areas to the elevated metal concentrations along the lower Dill and Lahn Rivers, I focus here on 9 tributaries whose headwaters are in the vicinity of former mines (Fig. 1). Four of the tributaries drain to the lower Lahn River and five to the lower Dill River. The earliest mention of mining in the region is 780 (Wenzel, 2003). Peak mining activity occurred between 1850 and the mid-1870s and from the early 1900s to the 1930s (Wenzel, 2003; Meyer, 2007). Following World War II, mining in the region declined, with the last mine (the Grube Fortuna iron mine) closing in 1983 (Wenzel, 2003; Meyer, 2007). Within the broad goal of assessing the contribution of formerly mined areas to trace metal concentrations along the lower Lahn and Dill Rivers, the specific objectives of this research are to (i) document metal concentration at near-channel sites along tributaries that drain formerly mined areas and along the lower Lahn River between Wetzlar and Weilburg, (ii) examine vertical trends in trace metal concentrations at sample sites, (iii) compare trace metal concentrations along tributaries to concentrations in the lower Lahn and Dill Rivers, and (iv) assess the contribution of tributaries to the elevated metal concentration along the lower Lahn and Dill Rivers.

2. Study area

The Dill River, with headwaters in the Lahn-Dill low mountains of west-central Germany, drains an area of ca. 700 km²; the Lahn River, with headwaters in the rural uplands north of Marburg, drains an area of ca. 5000 km² (Fig. 1). Unlike the Lahn River, the Dill River has not been extensively channelized, resulting in higher flow velocity and stream power than along the Lahn River. Overbank flow generally occurs during the winter and early spring (December to April), a period of higher precipitation and potential snowmelt from upland areas (Breuer et al., 2009). Slightly > 50% of the Dill River watershed is forested (mainly uplands), 27% is in agriculture (arable land and pasture), and 10% is urban (Fröhlich et al., 2008; Breuer et al., 2009). Land use in the Lahn River watershed is 70% agriculture (arable land and pasture), 16% forested, and 14% urban (Jöckel, 1981). Along both rivers, arable land is typically separated from the channel bank by a ca. 3-m-wide, near-channel vegetation zone of low-lying brush. The 9 tributary streams drain uplands adjacent to the lower Dill and Lahn Rivers. Tributaries have steeper altitudinal gradients compared to the gradients along the lower Dill and Lahn Rivers and flow through mainly forested areas. Along tributaries, slope gradients range from 0.44° to 2.12°, with the steeper gradients present along tributaries that flow into the lower Dill River; slope gradients along the lower Dill and Lahn Rivers range from 0.02° to 0.14°, with steepest gradients found along the lower Dill River.

Potential sources of trace metals documented in the lower Dill and Lahn Rivers include former steel mills, where trace metals were used as alloys in steel manufacture, in the upper part of the Dill River near and upriver from Herborn (Fröhlich et al., 2008), and the mines that were once active in uplands of the watershed (Meyer, 2007). Fe ore, but also Cu and Pb ore, were mined from Carboniferous shale; the same shale serves as a potential non-anthropogenic source of Cu, Pb, and Zn (Wetzel, 1994). Industrial activity represents an alternative source of trace metals in the basin. Analysis of 73 sites along a 60-km length of the Dill River revealed maximum concentrations of trace metals in stream water samples collected immediately downstream from steel mill locations (Fröhlich et al., 2008). The reach with the highest concentrations of trace metals corresponds to the lower Dill River from Herborn to Wetzlar where I documented anthropogenic storage of Cu, Pb, and Zn in near-channel floodplains (Martin, 2015).

3. Materials and methods

Through field and archival reconnaissance, I selected 9 tributary streams that drain formerly mined areas. With the exception of large mines like the Grube Fortuna (Fig. 2), evidence of mining activity from the generally small mines is often difficult to discern on the modern landscape, but is portrayed on 1:25,000 topographic maps. Samples were collected at 15 sites along tributary streams. Samples were also collected from 7 near-channel sites along the lower Lahn River between Wetzlar and Weilburg to better document trace metal concentrations there. Generally, sites were located at an interval of every 2–4 km along the lower Lahn River and every 1 km along tributary streams to ensure systematic coverage while avoiding potential sources of trace metals such as major roads, rail lines, and industrial facilities. Every attempt was made to select an equal number of left and right channel bank sites, although limited channel access made this difficult in a few reaches. The geographical latitude and longitude of each sample site was recorded using a hand-held GPS unit.

At each sample site, I collected a 2.5 cm diameter core using an Oakfield handheld corer. Cores were collected from near-channel sediments to a depth of at least 1.0 m along the lower Lahn and to a depth of ca. 0.6 m along tributaries. Coarse grained and rocky sub-surface materials prevented deeper coring at some tributary sites. In the field, cores were sampled at an interval of 5 cm for analysis of trace metals and standard sediment analyses. Samples were stored in sterile plastic

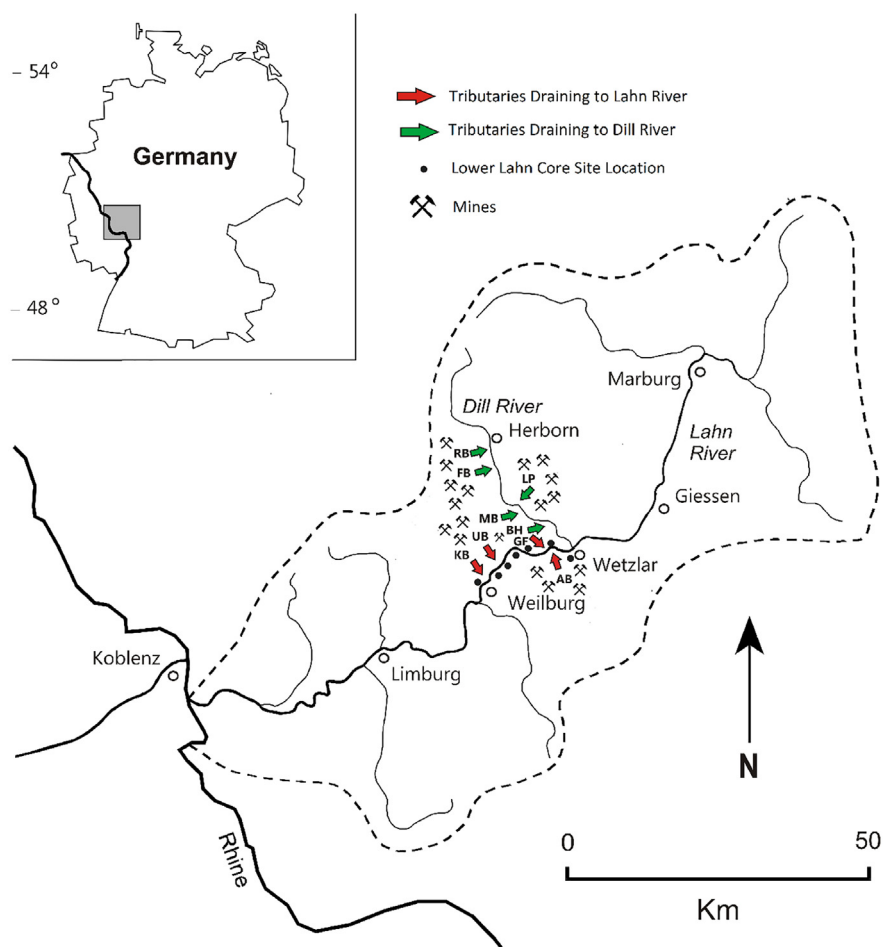


Fig. 1. Map of the study showing the location of former mining areas and tributaries draining to the lower Dill and Lahn Rivers. Two letter labels correspond to tributary stream names or geographic locations: AB - Albshausen; BH - Berghausen; FB - Fleisbach; GF - Grube Fortuna mine; KB - Kallenbach; LP - Lempbach; MB - Mühlbach; RB - Rehbach; UB - Ulmbach.



Fig. 2. Open mining pit at the Grube Fortuna mine. This pit was active from mid-1800s to WW 1; later mining activity was below ground. The mine is drained by tributaries GF and BH shown in Fig. 1.

bags and transported to the geomorphology lab at Justus-Liebig Universität in Giessen. In the lab, each sample was split, with half retained for grain size analysis and organic matter content determination and half submitted to a commercial laboratory for analysis of trace metals. At the lab, samples were sieved to separate the < 63 µm fraction on which trace metal analysis was completed. Trace metals are most often transported and stored on the < 63 µm fraction (Ntekim et al., 1993; Mantei and Sappington, 1994; Foster and Charlesworth, 1996; Owens et al., 2005); using this fraction also provides consistency with my previous research along the Lahn and Dill Rivers. Samples were first digested in perchloric, nitric, and hydrofluoric acids (triple-acid digestion), a process that has been found to yield nearly total recovery of trace metals (Lecce and Pavlowsky, 1997). The concentrations of Cu, Pb, and Zn were measured using inductively coupled plasma atomic emission spectroscopy (ICE-AES). The analytical sensitivity was 1.0 mg kg⁻¹ for Cu and 2.0 mg kg⁻¹ for Pb and Zn. Quality control information provided by the laboratory yielded coefficients of variation on duplicate samples of 2.3% for Cu, 3.7% for Pb, and 1.5% for Zn. Repeated measurements of standards by the lab produced coefficients of variation of 1.8% for Cu, 2.6% for Pb, and 2.0% for Zn. On the half of samples that I retained in the lab, the pipette method was used on 10 g samples to determine the proportions of clay (< 2 µm) and silt (2–62 µm); the percent sand was determined by wet sieving. Loss on ignition (LOI) on 3 g samples at 430° C for 24 h yielded organic matter content (Davies, 1974).

Deeply buried sediments were collected to provide background trace metal concentrations. Based on vertical metal trends and AMS radiocarbon ages on charcoal from sediment cores in the Dill River (Martin, 2015), sediments at depths > 0.6 m date to the pre- or early-industrial period (early 19th century and older), and are assumed to provide background trace metal concentrations in the study area. Background trace metal concentrations are derived from 27 background samples collected at depths between 0.60 and 1.4 m along the lower Lahn and Dill Rivers. Determining background (sometimes referred to as non-anthropogenic) trace metal concentrations can be problematic (Birch et al., 2000; Matschullat et al., 2000; Bindler et al., 2011). Matschullat et al. (2000) summarize and assess the options for determining background trace metal concentrations.

To assess the extent of metal contamination, trace metal concentrations from sediment cores were compared to standards contained in the 1998 German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz* {BbodSchG}) and the German Federal Soil Protection and Contaminated Site Ordinance of 1999 (*Bundes-Bodenschutzverordnung*{BbodSchV}). These federal acts set trace metal concentrations above which the “natural soil functions” are thought to be at risk (the ‘precautionary value’). Values are reported for loam/silt soils and are given in mg kg⁻¹ dry mass fine soil. In addition to standards provided in the German Federal Protection Act, I compared trace metal concentrations to the threshold effect concentrations (TEC) and probable effect concentrations (PEC) of MacDonald et al. (2000). When concentrations are less than the TEC, adverse impacts are unlikely; with concentrations above the PEC, negative impacts are deemed likely.

4. Results

4.1. Summary metal concentrations

The descriptive statistics for core samples and background samples from tributaries, the lower Lahn River, and the lower Dill River are summarized in Table 1. Results from a Mann-Whitney *U* Test indicate that the median concentrations of Cu, Pb, and Zn from tributaries to both the lower Lahn and Dill Rivers are statistically different than median concentrations for the respective rivers into which the tributaries empty. Mean concentrations along the tributaries are virtually identical to background concentrations for the lower Lahn and Dill Rivers. Mean concentrations of Cu, Pb, and Zn are far less than the

Table 1

Summary trace metal concentrations along tributaries, the Lower Lahn River, and Dill River. All values are for dry mass.

	Cu (mg kg ⁻¹)	Pb (mg kg ⁻¹)	Zn (mg kg ⁻¹)
Lahn tributaries (n = 69)			
Mean	20.9	26.7	77.9
Median	20.0^a	25.0	74.0
SD ^b	6.6	6.2	17.9
Range	13–38	17–45	48–125
Dill tributaries (n = 72)			
Mean	29.9	31.3	98.8
Median	26.0	31.0	90.5
SD	12.3	9.0	32.6
Range	15–65	17–54	52–179
Lower Lahn River (n = 165)			
Mean	64.2	95.4	318.2
Median	67.0	92.0	349.0
SD	31.4	49.8	160.9
Range	18–152	19–232	63–741
Lower Dill River ^c (n = 213)			
Mean	83.4	107.9	323.9
Median	80.0	99.0	345.0
SD	41.1	73.4	179.8
Range	25–229	20–408	73–992
Background - Lower Lahn (n = 8)			
Mean	30.5	47.6	128.5
Median	24.0	36.0	94.0
SD	15.4	31.6	78.5
Range	17–59	20–108	69–262
Background - Lower Dill (n = 19)			
Mean	37.5	39.3	105.0
Median	35.0	34.0	99.0
SD	9.9	17.6	30.1
Range	25–57	20–77	73–169
Precautionary Value ^d	40.0	70.0	150.0
TEC ^d	31.6	35.8	121.0
PEC ^d	149.0	128.0	459.0

^a Median concentrations for tributaries shown in bold are statistically different ($p \leq 0.001$) than median concentrations for the river into which each drains.

^b SD - Standard Deviation.

^c Lower Dill River data from Martin (2015).

^d The “Precautionary Value” from the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz*) of 1998; TEC (threshold effect concentration) and PEC (probable effect concentration) values from MacDonald et al. (2000).

precautionary value contained in the German Federal Soil Protection and Contaminated Site Ordinance of 1999 and the TEC and PEC levels of MacDonald et al. (2000). Along tributaries that drain into the Dill River, a few individual samples (as reflected in the range of concentrations) exceed the precautionary and TEC values for Cu and Zn and the TEC value for Pb. No individual samples along tributaries that drain into the lower Lahn River exceed the TEC, PEC, or precautionary values for any of the three trace metals.

The mean concentrations of Cu, Pb, and Zn along the lower Lahn River are greater than background concentrations (Table 1). Mean concentrations exceed precautionary values and TEC values for all three trace metals, but are less than the PEC values. There are individual samples along the lower Lahn River whose trace metal concentrations are greater than the PEC value for all three metals. As shown in Table 2, the concentrations of Cu, Pb, and Zn in near-channel sediments of the lower Lahn River valley are greater than near-channel concentrations along the upper Lahn River valley (upstream from Wetzlar). With the exception of mean Pb concentrations in the upper Lahn River, the mean concentrations of all three trace metals in both reaches exceed precautionary values and the TEC. There is anthropogenic enrichment of Cu, Pb, and Zn in both the upper and lower Lahn River, but the

Table 2

Summary trace metal concentrations along the Upper and Lower Lahn River. All values are for dry mass.

	Cu (mg kg ⁻¹)	Pb (mg kg ⁻¹)	Zn (mg kg ⁻¹)
Upper Lahn ^a (n = 145)			
Mean	46.8	65.8	229.4
Median	43.0^b	60.0	218.0
SD ^c	13.6	21.8	67.1
Range	21–94	32–196	86–406
Lower Lahn (n = 165)			
Mean	64.2	95.4	318.2
Median	67.0	92.0	349.0
SD	31.4	49.8	160.9
Range	18–152	19–232	63–741
Background - Lahn River (n = 42)			
Mean	24.4	30.9	93.2
Median	20.5	21.0	72.0
SD	13.5	26.2	54.1
Range	11–82	12–136	42–262
Precautionary			
Value	40.0	70.0	150.0
TEC	31.6	35.8	121.0
PEC	149.0	128.0	459.0

^a Upper Lahn River data from Martin (1997, 2004, 2012).

^b Median concentrations that are statistically different between the Upper Lahn and Lower Lahn are shown in bold ($p \leq 0.001$).

^c SD - Standard Deviation.

magnitude of enrichment is greater in the lower Lahn. In addition, the most contaminated individual sites, as indicated by the maximum concentrations for each trace metal, are present in the lower Lahn River valley.

4.2. Vertical trace metal trends

Vertical metal trends from individual cores reflect the quantity of contaminated sediments that have moved through and been deposited along rivers. Although post-depositional translocation of metals is possible, especially for those that are less strongly bound to sediments (Hudson-Edwards et al., 1999), similar “down-core trends” for different trace metals indicate minimal post-depositional mobilization (Lecce and Pavlowsky 2014, 126–128). Vertical trends in trace metals and percent silt and clay from representative cores collected along tributary streams are shown in Fig. 3; vertical trends from other tributary cores are shown in Supplementary materials. Two letter designations refer to the tributary name (e.g., LP-Lempbach) or nearby geographic location (e.g., GF - Grube Fortuna mine). Numbers reflect the position along the tributary, with higher numbers positioned farther downstream. Peaks in trace metal concentrations are present at specific depths below the surface at several tributary sites, but are less pronounced than along the lower Dill River (Martin, 2015). In addition, the range of metal concentrations along tributaries is far less than the range from the lower Dill River (Table 1). The depth of peak concentrations varies widely from 0 to 0.45 m for Cu, Pb, and Zn. For those cores where the peak concentration occurs at a single depth, the mean depths of the peak concentrations (and standard deviations) for Cu, Pb, and Zn are 0.20 (0.17), 0.12 (0.15), and 0.06 m (0.09), respectively ($n = 10$). Vertical trends for percent silt and clay reveal no relationship between sediment grain size and trace metal concentrations.

In contrast to vertical metal trends noted along tributary streams, the trends in cores collected along the lower Lahn River (Fig. 4) are similar to trends previously reported along the Dill River (Martin, 2015); additional vertical trends are shown in Supplementary materials. Most cores along the lower Lahn River display a pronounced single peak in metal concentrations at depths between 0.2 and 0.5 m. In individual cores, the vertical trends for the three trace metals are generally similar, suggesting minimal post-depositional migration of

metals. For those cores along the lower Lahn River where the peak concentration occurs at a single depth, the mean depths of the peak concentrations (and standard deviations) for Cu, Pb, and Zn are 0.29 (0.09), 0.33 (0.17), and 0.29 m (0.09), respectively ($n = 5$), similar to what was noted along the lower Dill River (Martin, 2015). Along the lower Lahn River, there is no systematic difference in vertical trends and trace metal concentrations with distance downstream from the confluence of the Dill and Lahn Rivers in Wetzlar (Fig. 4). Sediments deposited most recently (in the upper 0.40 m of cores) are less contaminated with trace metals than sediments lower in the cores, an indication that cleaner sediments have begun to move through the lower Lahn River during the last several decades. Similar declines in metal concentrations in floodplain sediments have been noted elsewhere in western Europe (e.g., Brüggemann, 1995; Zerling et al., 2006; Zachmann et al., 2013; Schillereff et al., 2016).

4.3. Tributary stream metal concentrations

The concentrations of trace metals in near-channel positions along tributary streams are less than concentrations along the lower Dill and Lahn Rivers. Mean concentrations along the five Dill tributaries for Cu, Pb, and Zn are 36%, 29%, and 31%, respectively, of the means for those trace metals along the lower Dill River; mean concentrations along the four Lahn tributaries for Cu, Pb, and Zn are 33%, 28%, and 24%, respectively, of the means for those trace metals along the lower Lahn River. The maximum values for the three trace metals noted along the five tributary streams to the Dill River range from 17% (Zn) to 25% (Cu) of the maximum concentrations noted along the lower Dill River; maximum values for the three trace metals noted along four tributary streams to the Lahn River range from 13% (Pb) to 28% (Cu) of the maximum concentrations noted along the lower Lahn River. Box plots of the median and mean values for the tributary streams vs. the lower Dill and Lahn Rivers illustrate the difference in metal concentrations (Fig. 5).

5. Discussion

Cores collected along tributary streams suggest that large quantities of Cu, Pb, and Zn from upstream mining activity are not present in near-channel positions. Some vertical variation in trace metal concentrations exists, but differences between peak concentrations and minimum values are small. Given that trace metals are often found in association with fine-grained sediments (Horowitz, 1985; Foster and Charlesworth, 1996; Owens et al., 2005), the low metal concentrations could be explained by absence of fine-grained sediments along the relatively steep-gradient tributary streams. A lack of fine-grained sediments along tributaries is not, however, supported by grain-size data from 56 samples collected along the tributaries. These samples reveal that sediments are predominantly fine-grained, with the mean percentage of silt and clay at 86.9% (standard deviation of 9.2%). Rather than being a function of the grain size of sediments, the low concentrations of Cu, Pb, and Zn along tributaries indicate that metal-contaminated sediments are not present.

The absence of high concentrations of trace metals along tributaries is surprising given the long history of Fe, Cu, and Pb mining in the study area. Perhaps the mining process did not release significant quantities of metals to tributaries, although studies of mined watersheds elsewhere in Germany (e.g., Matschullat et al., 1997; Monna et al., 2000; Hürkamp et al., 2009) and western Europe (e.g., Coulthard and Macklin, 2003; Beane et al., 2016; Bird, 2016) found significant levels of metals stored along rivers immediately downstream from mines. Contaminated sediments may have been conveyed through, rather than stored along, the relatively high gradient tributaries to the lower Dill and Lahn Rivers; gradients along tributary streams are much greater than gradients along the lower Dill and Lahn Rivers, although the maximum gradient found along tributaries to the Dill (2.12°) is still

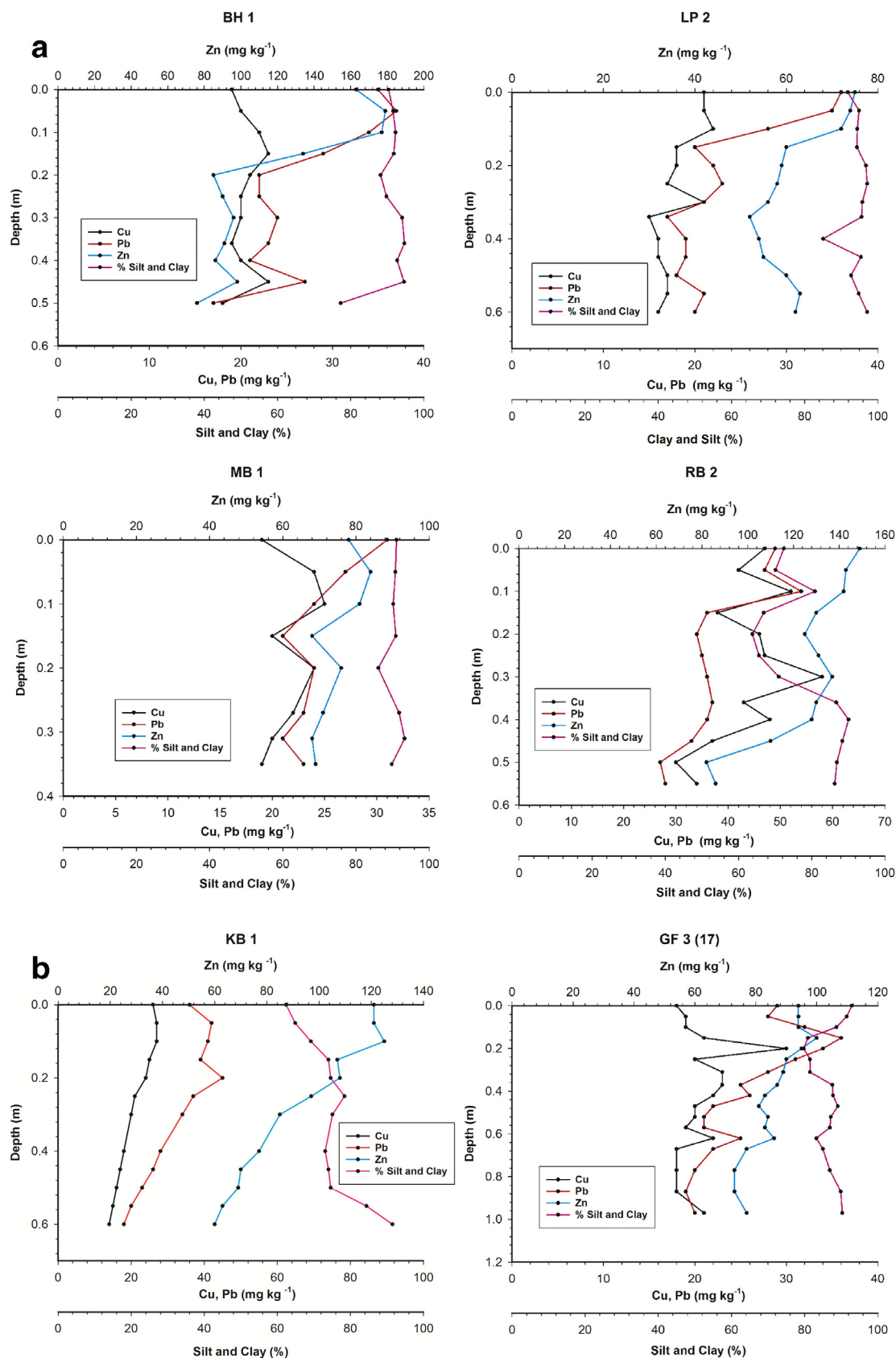


Fig. 3. Vertical trends in Cu, Pb, and Zn and percentage silt and clay at representative core sites along tributary streams to the (a) lower Dill River and (b) lower Lahn River.

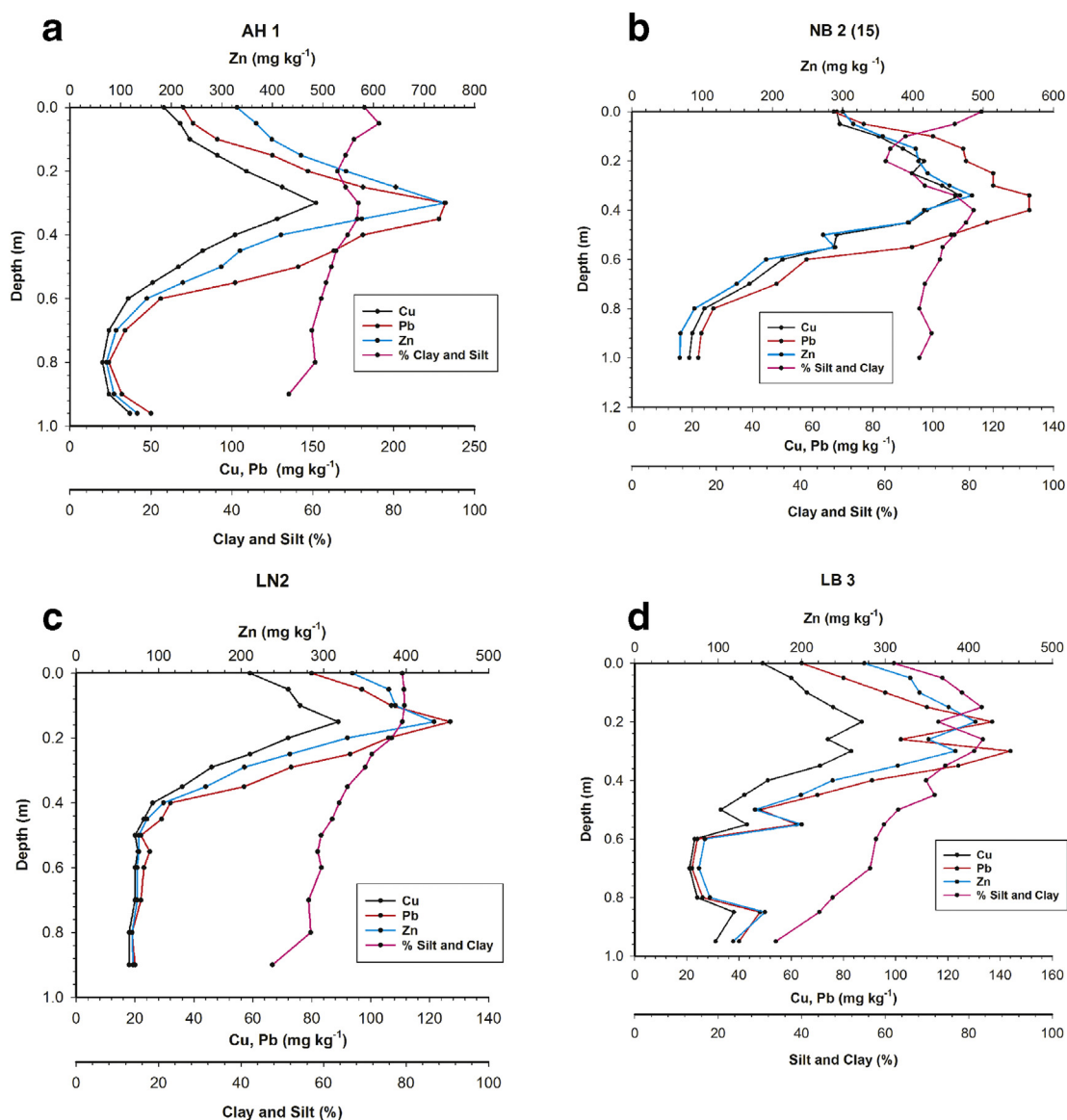


Fig. 4. Vertical trends in Cu, Pb, and Zn and percentage silt and clay for cores collected along the lower Lahn River. Along the lower Lahn River, site (a) is the most upstream location and site (d) the most downstream location. Two letter labels correspond to nearby geographic locations (e.g. LB refers to the village of Löhnberg).

relatively low. If metals had been moved through rather than deposited along tributary streams, relatively high concentrations of metals might be expected in cores collected close to the mouth of tributaries where stream gradient, and thus stream power, decrease. In addition, tributary floodplains are wider in the lower reaches, which would favor deposition. Samples taken from the most downstream sites along tributaries (Fig. 3 - GF3(17), LP2) do not have higher metal concentrations than samples collected upstream, suggesting that higher concentrations of metals were either 1) released to tributaries and then conveyed directly to the main channel, or 2) never transported through tributary streams.

If metals had been released to tributaries, but subsequently transported downstream through them to the main valleys, cores collected along floodplains of the lower Dill and Lahn Rivers immediately downstream of tributary confluences with the main channel might be expected to have higher concentrations of trace metals than floodplains immediately upstream of confluences. The maximum and mean metal concentrations for each sample site along the lower Dill and Lahn Rivers and the locations where each tributary enters the main channel do not consistently reveal higher concentrations immediately downstream from tributary confluences (Fig. 6). The spatial pattern of trace

metals along the main channel floodplains of the lower Dill and Lahn Rivers is not consistent with tributaries contributing large quantities of trace metals.

It appears that mined areas have not served as an important source for anthropogenic metals found along the lower Dill and Lahn Rivers, perhaps because mining did not release large quantities of metals to the fluvial environment. Cores show minimal trace metal concentrations in tributary floodplain sediments, meaning there is little risk that sediments stored along tributaries will serve as a future source of downstream trace metal contamination. Instead, elevated anthropogenic concentrations of metals along the lower Lahn River appear to have originated from industrial sources such as those noted in the upper Dill River by Fröhlich et al. (2008). The peak metal concentrations in cores collected along the lower Lahn River are assumed to correspond to, or directly follow, maximum metal releases to the environment. Industrial activity in the upper Dill River is ongoing, but the peak period of activity, and likely period of less restrictive environmental regulations, occurred from the mid-1800s to the mid-1900s (Fröhlich et al., 2008). This timeline of the expansion and contraction of industrial activity has been noted elsewhere in studies that document the input of trace metals

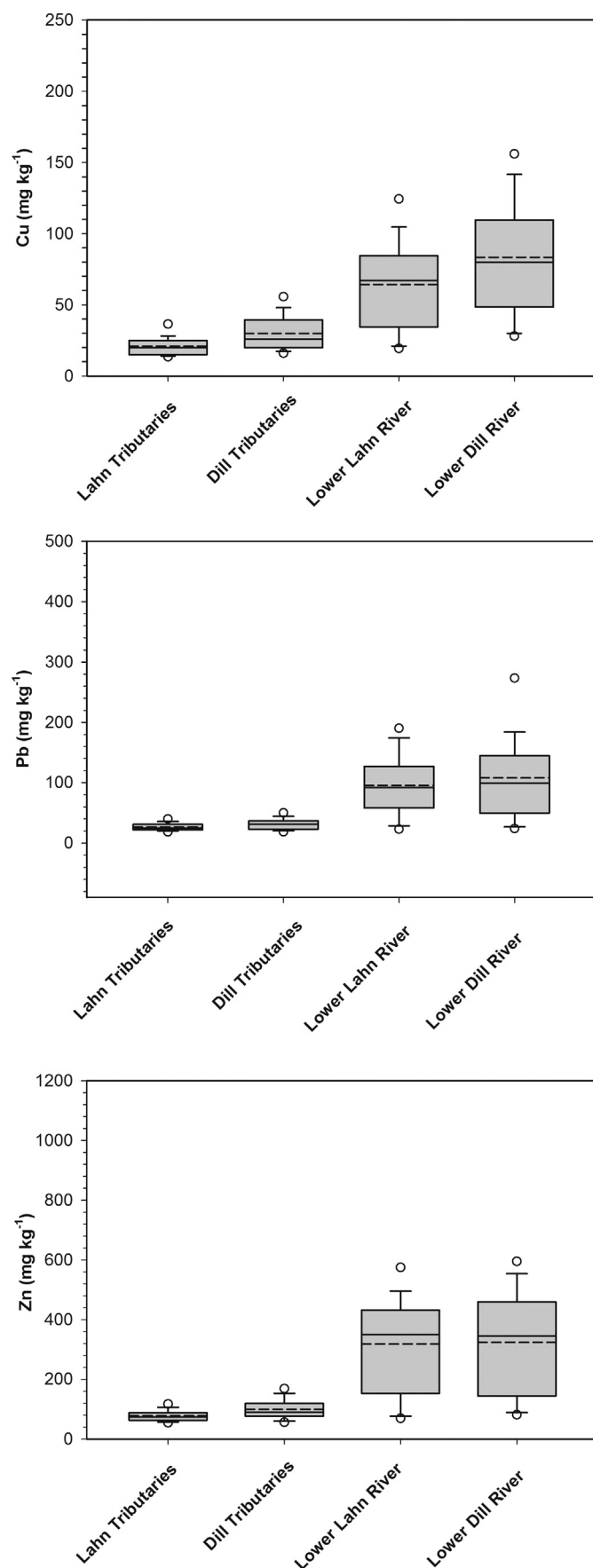


Fig. 5. Box plots depicting median concentrations (solid line) and mean concentrations (dashed line) of Cu, Pb, and Zn for tributary streams vs. lower Dill and Lahn Rivers. 25th and 75th percentiles shown within the box. Error bars display the 10th and 90th percentiles; open circles are the 5th and 95th percentile values.

to watersheds (Valette-Silver, 1993; Matschullat and Bozau, 1996; Schulte et al., 1996; Renberg et al., 2001; De Vleeschouwer et al., 2007; Baborowski and Einax, 2016; Dhivert et al., 2016; Horák and Hejeman, 2016).

AMS radiocarbon ages on charcoal from alluvium along the lower Dill River indicate that sediments with peak concentrations of metals were deposited after the mid/late seventeenth century, with several sites displaying peak metal concentrations at 100 to 150 years ago (Martin, 2015). Assuming the alluvial stratigraphy along the lower Lahn River is coeval with that along the lower Dill River, the peak metal concentrations at ca. 0.30 m date to the late 1800s and early 1900s. The releases of pollutants by industrial activities to the environment has lessened over recent decades as indicated by vertical trends from the lower Dill (Martin, 2015) and Lahn Rivers (Fig. 4) showing up-core declines in metal concentrations. Although recently-deposited sediments are cleaner, near-channel sediments along the lower Dill and Lahn Rivers still contain metal-contaminated sediments that may serve as potential sources of downstream trace metal pollution in the foreseeable future as has been noted in other western European watersheds (e.g., Förstner, 2003; Krüger et al., 2005; Macklin et al., 2006; Foulds et al., 2014; Resongles et al., 2014).

6. Conclusions

A century of mining in uplands adjacent to the lower Dill and Lahn Rivers of central Germany does not appear to have contributed trace metals to floodplain sediments of tributaries. Concentrations of Cu, Pb, and Zn in cores collected from 9 tributary streams that empty into the lower Dill River or Lahn River differ little from background metal concentrations and are significantly less than trace metal concentrations from near-channel sediments along the lower Dill and Lahn Rivers. Mean concentrations of Cu, Pb, and Zn are below the precautionary value contained in the German Federal Soil Protection Act and the TEC and PEC values of MacDonald et al. (2000), although some individual samples exceed the precautionary and TEC values. Vertical metal trends in tributary cores reveal peak concentrations at depth below the surface, but peak concentrations differ little from minimum concentrations in the majority of cores, typically by a factor of only 1.5 to 2.0. Peak concentrations in tributary cores are far less than peak concentrations in cores extracted from the lower Dill River and lower Lahn River. The metals along the lower Lahn and Dill Rivers appear to have been derived from upstream industrial activity in cities such as Herborn and Dillenburg (Fröhlich et al., 2008).

This study serves as an example of an historically mined area not contributing to anthropogenic concentrations of trace metals found in downstream near-channel locations. Perhaps because the mines were generally small, the mining and processing of ore did not leave large quantities of contaminated materials exposed on the landscape where they could be transported to watersheds. It appears that the former mines and streams that drain them have a very low potential of serving as sources for metals to downstream rivers in the near future. Downstream water and floodplain soil quality do not appear to be endangered by the contribution of trace metals from upland mined areas. At the same time, relatively high concentrations of Cu, Pb, and Zn are stored in near-channel positions along the lower Lahn River as well as along the lower Dill River. Channel bank erosion has the potential of entraining these metal-contaminated sediments and transporting them downstream. Given the high concentrations of metals present along the lower Dill River, the lower Lahn River into which the Dill empties at

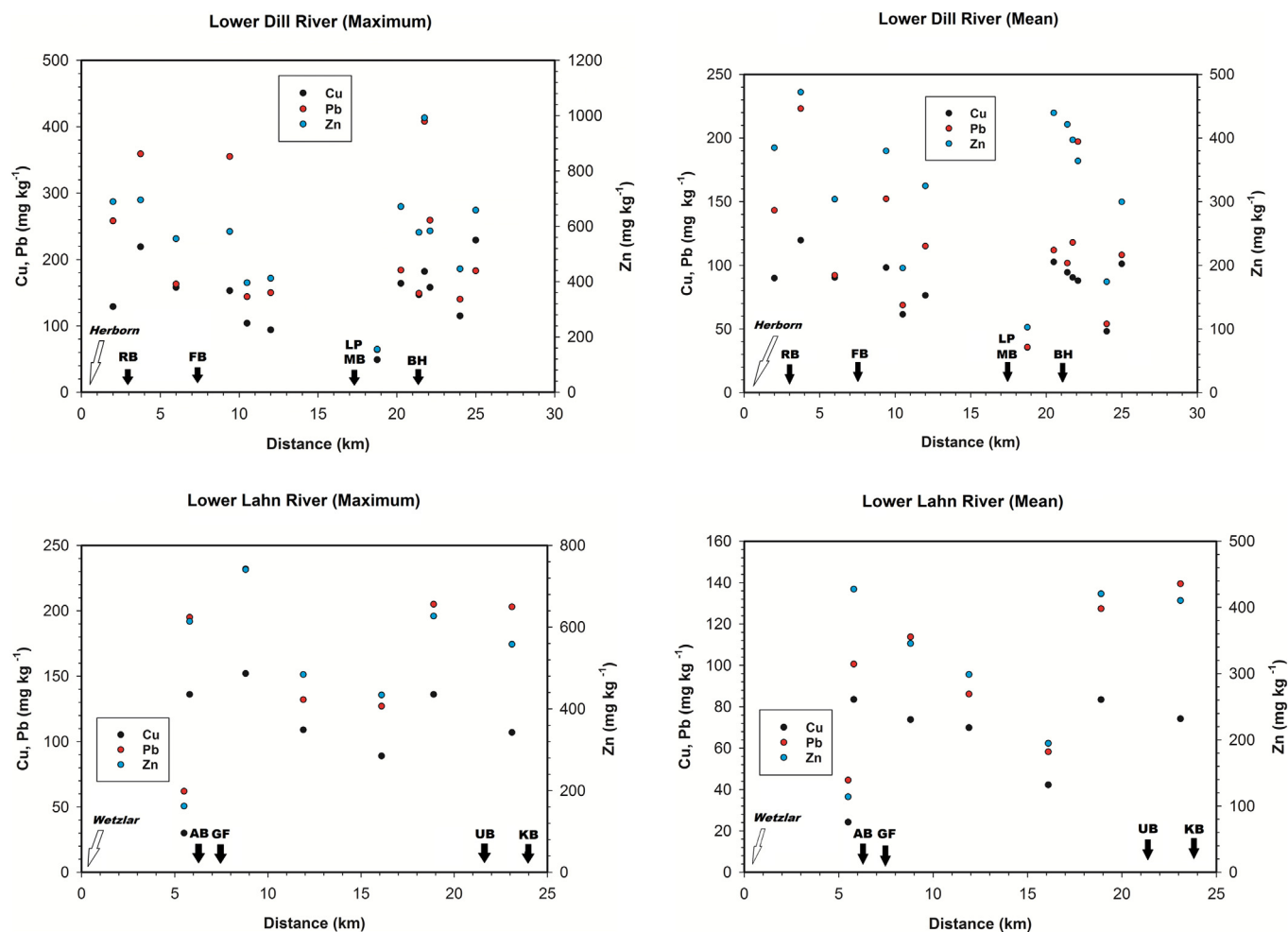


Fig. 6. Maximum and mean concentrations for each sample core along the lower Dill and Lahn Rivers. Arrows show locations where tributary streams empty into main channel; two letter labels correspond to tributary stream names. Distance measured downstream from Herbörn (lower Dill River) and Wetzlar (lower Lahn River).

Wetzlar could continue to receive metal-contaminated sediments for the foreseeable future.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.catena.2018.11.008>.

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